

RockMauler™ CR-220 Excavation Crusher

OEM Field Manual — Volume 2

AI-Augmented Field Engineer Edition

SECTION 1 — SAFETY & COMPLIANCE

1.1 General Safety

- Only certified mining/construction personnel may operate or service the CR-220.
- Always perform Lockout/Tagout before accessing the crusher chamber or rotor assembly.
- Maintain a minimum 4 m clearance around the feed hopper during operation.
- Wear Class-IV PPE: reinforced gloves, dust respirator, anti-vibration boots.

1.2 Hazard Zones

- **Red Zone:** Crusher chamber (crush hazard, high-impact surfaces).
- **Orange Zone:** Rotor assembly (entanglement hazard).
- **Yellow Zone:** Feed hopper (fall-in hazard).

1.3 AI-Assisted Safety

The AI Field Engineer Tool automatically:

- Detects chamber pressure spikes indicating imminent jams.
 - Predicts rotor overload conditions using torque harmonics.
 - Flags unsafe feed patterns (oversized rocks, uneven loading).
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SECTION 2 — SYSTEM OVERVIEW

2.1 Machine Description

The RockMauler™ CR-220 is a mid-capacity excavation crusher designed for breaking down rock, ore, and construction debris directly at the job site. Its dual-jaw system and rotor assembly provide high throughput with minimal downtime.

2.2 Core Subsystems

- **Crushing System:** Dual jaw plates + RA-220 rotor
- **Powertrain:** MU-18 motor + BD-Flex belt drive
- **Control System:** Shock-resistant operator console
- **Sensor Array:** Chamber pressure, rotor speed, jam detection

SECTION 3 — BILL OF MATERIALS (BOM)

3.1 Structural Assembly

- Crusher Housing (Impact-Grade Steel)
- Feed Hopper (Anti-Abrasion Lining)
- Discharge Chute (Vibro-Assist)

3.2 Crushing System

- Dual Jaw Plates (Titanium-Carbide)
- Rotor Assembly RA-220
- Chamber Alignment Rails
- Jam-Release Actuator (JRA-10)

3.3 Powertrain

- Motor Unit MU-18 (18 kW)
- Belt Drive System (BD-Flex)
- Cooling Fan Assembly (CF-220)

3.4 Sensor & Control

- Chamber Pressure Sensors (CPS-3)
- Rotor Speed Sensor (RSS-5)
- Jam Detection Array (JDA-2)
- Operator Console (Shock-Resistant)

3.5 Consumables

- Jaw Plate Wear Inserts
- Belt Lubricant
- Dust Filter Pads

SECTION 4 — OPERATING PROCEDURES

4.1 Startup Sequence

1. Inspect feed hopper for debris.
2. Power on MU-18 motor unit.
3. Run **AI-Assisted Rotor Calibration** (8 seconds).
4. Activate belt drive system.
5. Begin controlled feeding of material.

4.2 Crushing Workflow

- Maintain rotor speed between 1,200–1,450 RPM.
 - AI flags oversized material before chamber entry.
 - Use Vibro-Assist discharge mode for wet or clay-heavy loads.
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SECTION 5 — FIELD DIAGNOSTICS

5.1 Common Failures

- Chamber jams
- Rotor torque overload
- Belt slippage
- Jaw plate misalignment

5.2 AI-Assisted Diagnostics

The AI tool provides:

- Chamber pressure spike graphs
- Rotor torque harmonic analysis
- Belt tension prediction
- Jaw alignment deviation mapping

5.3 Field Repair Procedure: Clearing a Chamber Jam

1. Activate **AI Jam Scan**.
 2. AI identifies jam type (material blockage, misaligned jaw, rotor obstruction).
 3. Engage Jam-Release Actuator (JRA-10).
 4. Remove obstructing material manually if safe.
 5. Inspect jaw plates for wear >35%.
 6. Re-run jam scan to confirm chamber clearance.
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SECTION 6 — MAINTENANCE

6.1 Daily Maintenance

- Clean feed hopper lining.
- Inspect jaw plate wear.
- Check belt tension.

6.2 Weekly Maintenance

- Replace dust filter pads.
- Lubricate belt drive system.
- Run AI-guided **Rotor Health Scan**.

6.3 Monthly Maintenance

- Replace jaw plate wear inserts.
- Perform full chamber alignment.
- Update firmware via AI Fleet Console.

SECTION 7 — TROUBLESHOOTING MATRIX

Issue	Likely Cause	AI Diagnostic Output	Field Fix
Chamber Jam	Oversized material	Pressure spike graph	Clear jam, inspect jaws
Rotor Overload	Material density spike	Torque harmonic distortion	Reduce feed rate
Belt Slippage	Low tension	Belt tension warning	Adjust BD-Flex tension
Jaw Misalignment	Wear or impact	Alignment deviation map	Replace inserts

SECTION 8 — TECHNICAL SPECIFICATIONS

- **Rotor Speed:** 1,450 RPM max
- **Jaw Force:** 22,000 N
- **Motor Power:** 18 kW
- **Throughput:** 18–24 tons/hr
- **Operating Temp:** –15°C to 60°C
- **Weight:** 6,800 kg